AI-Augmented Research

Session 1.1: Research Judgement in an AI Environment

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instats Seminar

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Interactive App and Repository

- **Streamlit App URL:** Click here to open the Interactive Streamlit App
- **No Installation:** Runs entirely in the browser via Streamlit — no Python or programming knowledge required.
- **Navigation:** Use the sidebar to switch between the six seminar sessions across the three days.
- **Step-by-Step:** Each page walks through worked examples and a Bring Your Own Data (BYOD) interface for that session's core workflows — from research question development and literature scoping to data quality auditing, end-to-end computational pipelines, manuscript integrity checks, and AI disclosure evaluation.
- **Consistency:** Results match the companion Python scripts and prompt library available in the `scripts/` folder of the seminar GitHub repository.
- **GitHub Page URL:** Click here to open the seminar GitHub page

Day 1 Overview

1. How AI Is Changing Research Practice
2. Understanding the Current AI Ecosystem
3. The Labour-versus-Judgement Line
4. Thinking with AI as a Cognitive Partner
5. Research Question Development
6. The Zyphur Framework

AI Is Reshaping the Research Lifecycle at Every Stage

AI is no longer confined to data analysis. It now touches every stage of the research lifecycle: question formation, literature discovery, data collection, analysis, writing, and peer review. The question is no longer whether AI will be part of research, but how researchers should engage with it responsibly.

- HAI Stanford (2024): 17.5% of CS papers and 16.9% of peer review text had AI-drafted content
- *Nature* (2026): tens of thousands of 2025 publications may contain invalid AI-generated references
- Zhang et al. (2025, *npj Complexity*): LLMs are redefining the scientific method across disciplines

Organizing question

What should the researcher decide, and what can AI safely assist with?

The AI Ecosystem: Not a Single Tool but a Family of Systems

Researchers often ask “Should I use ChatGPT?” The better question is: **which type of system is appropriate for this task?**

System Type	Strengths	Research Use Cases	Key Limitations
General LLMs	Broad language tasks	Writing, brainstorming	Hallucination, sycophancy
Reasoning models	Multi-step logic	Method selection, critique	Slower, higher cost
Retrieval-augmented (RAG)	Grounded in sources	Literature synthesis	Quality depends on corpus
Multimodal systems	Text + image/audio	PDF extraction	Variable accuracy
AI agents	Autonomous multi-step	Data collection, analysis	Reproducibility risk
Local/open-source	Privacy, auditability	Sensitive data research	Lower capability

Choosing the Right Tool Is Itself a Research Decision

Tool selection is not a technical afterthought; it is a **methodological choice** with consequences for validity, reproducibility, and transparency.

Zyphur (2026) argues that AI literacy is fundamentally a **judgement problem**, not a tooling-skills problem. The researcher who selects a tool without understanding its failure modes has made an undisclosed methodological decision.

- Zyphur (2026): Dimension 2 — four AI-use modes (Search, Co-author, Validator, Tutor)
- Blau et al. (2024, *PNAS*): transparent disclosure and attribution as the first principle of scientific integrity

The Labour-versus-Judgement Line: The Seminar's Organizing Principle

Zyphur (2026) draws a fundamental distinction between two categories of research task.

Labour tasks

Can be safely augmented by AI with modest verification cost:

- Formatting and typesetting
- Transcription
- Bibliography management
- Code boilerplate

Judgement tasks

Must remain human:

- Framing the research question
- Interpreting an unexpected result
- Weighing competing evidence
- Deciding what to publish

Misclassifying a judgement task as a labour task is the most common error in AI-augmented research practice (Zyphur, 2026).

Worked Examples: Labour versus Judgement

Task	Classification	Rationale
Formatting citations to APA style	Labour	Rule-based, verifiable
Transcribing an interview recording	Labour	Verifiable against audio
Summarizing a paper you have read	Labour (with verification)	Must check against source
Framing the research question	Judgement	Defines scope of inquiry
Interpreting a null result	Judgement	Requires domain expertise
Deciding which covariates to include	Judgement	Analytic choice with consequences
Evaluating a reviewer's critique	Judgement	Requires scholarly authority

The Discovery-versus-Verification Distinction

A second organizing distinction runs through all three days of this seminar.

Discovery

Where AI adds the most speed and breadth:

- Generating possibilities
- Exploring literature
- Proposing methods
- Surfacing analogies

Verification

Where human judgement is most critical:

- Checking references
- Validating claims
- Reproducing computations
- Finding inconsistencies

Conflating the two — treating AI-generated outputs as verified findings — is the root of most AI-related research integrity failures.

AI as a Cognitive Partner: Beyond "Idea Generation"

The conventional framing of AI as an "idea generator" understates its potential and misrepresents its risks. A more accurate framing is **AI as a cognitive partner** — a system that can extend the researcher's thinking in structured ways.

- Wieland (2022, *Frontiers in AI*): chatbot brainstorming produces more ideas with greater diversity than human-only brainstorming
- Memmert et al. (2025): cognitive stimulation and free-riding in human-AI group dynamics
- Dockterman (2026, Harvard): cognitive partnering risks reinforcing group think without critical analysis

This reframing implies both richer uses *and* more demanding verification disciplines.

Five Modes of Cognitive Partnership

Mode	What the researcher does	What AI does	Verification discipline
Brainstorming	Provides research context	Generates candidate ideas	Evaluate on merits, not AI confidence
Analogy-finding	Specifies source domain	Proposes cross-disciplinary analogies	Check structural validity
Assumption-challenging	States working hypothesis	Identifies unstated assumptions	Decide which are substantive
Devil's advocate	Presents the argument	Constructs strongest counterargument	Assess whether counter is valid
Hypothesis refinement	Proposes initial hypothesis	Suggests precision and testability	Researcher retains final authority

Sycophancy: The Cognitive Partnership Failure Mode

The most dangerous failure mode in cognitive partnership is not hallucination but **sycophancy**: AI systems that agree with the user's framing, reinforce existing beliefs, and produce outputs that feel validating but lack objectivity.

- Cheng et al. (2026): sycophantic AI decreases prosocial intentions and promotes self-serving behavior
- PMC (2026): algorithmic sycophancy is a new source of systematic distortion in AI-assisted research
- Stanford Report (2026): chatbots are overly agreeable when giving interpersonal advice, affirming behavior even when harmful

Zyphur (2026) Competency 5

AI agreement is not evidence. Flat agreement with the researcher's framing is a failure signal, not a validation signal.

Detecting Sycophancy in Practice

Three practical detection strategies:

1. **Adversarial prompting:** explicitly instruct the AI to argue against your position and evaluate whether the counter-arguments are substantive or superficial
2. **Model heterogeneity:** run the same question through a model from a different AI family; persistent agreement across families is stronger evidence than single-model agreement
3. **Confidence calibration:** ask the AI to rate its own confidence and explain the basis; uncalibrated high confidence on contested questions is a sycophancy signal

Zyphur (2026): Competency 5 — human-as-verifier discipline; Competency 4 — model-heterogeneity in adversarial review.

Research Question Development: What AI Can and Cannot Do

AI can assist with:

- Scoping the question (what has already been asked)
- Sharpening the question (improving precision and testability)
- Surfacing related questions (adjacent literatures)

AI cannot:

- Determine whether the question is worth asking (contribution)
- Assess whether the question is feasible (resources)
- Decide what the answer means for theory or practice (significance)

The researcher's intellectual authority begins at the question.

Prompt Discipline: Prompts as Researcher Degrees of Freedom

The choice of prompt is a **researcher degree of freedom** analogous to analytic choice in the p-hacking literature. Different prompts to the same model on the same question can produce substantively different outputs.

- Simmons, Nelson & Simonsohn (2011, *Psychological Science*): researcher degrees of freedom and false-positive psychology
- Zyphur (2026): Competency 3 — prompt selection as a methodological choice requiring documentation

Documentation standard for every AI-assisted step:

1. Exact prompt text (verbatim)
2. Model name and version
3. Temperature and generation parameters
4. Date of the query
5. Full output (or representative sample)
6. Verification step and its result

The Zyphur Framework: Five Dimensions of Responsible AI Use

Zyphur (2026) organizes responsible AI use in research around five dimensions:

1. **The human-in-the-loop demarcation** — labour vs. judgement
2. **Responsible use in practice** — four AI-use modes
3. **Tooling that promotes responsible use** — six observable properties
4. **AI-literate humans** — six core competencies
5. **Institutional benchmarking** — four axes of readiness

Today's sessions cover Dimensions 1, 2, and 4. Day 2 covers Dimensions 3 and 4. Day 3 covers Dimension 5.

The Four AI-Use Modes (Zyphur Dimension 2)

Mode	Description	Primary verification discipline
Search	AI as retrieval and scoping tool	Every citation checked against primary source
Co-author	AI as writing and revision assistant	Researcher can independently produce and defend content
Validator	AI as critic and adversarial reviewer	Run through model from different AI family
Tutor	AI as structured learning partner	Corpus-grounded retrieval preferred over open chat

Zyphur (2026), Dimension 2.

Six Core AI-Era Competencies (Zyphur Dimension 4)

1. **Citation verification** — check every AI-generated reference against the primary source
2. **Model and parameter specification** — report model, version, temperature, seed, re-run stability
3. **Prompt discipline** — document prompts as researcher degrees of freedom
4. **Model-heterogeneity in adversarial review** — run critique through a different AI family
5. **Sycophancy detection and human-as-verifier discipline** — AI agreement is not evidence
6. **Structured failure-mode reporting** — report protocol, threat model, and failure-discovery rate

Zyphur (2026), Dimension 4. These competencies are not Day 3 topics; they appear in every session from Day 1 onward.

Session 1.1 Summary

Three things to carry forward

1. **Research judgement** — the question of what the researcher must decide and what AI can safely assist with — is the organizing thread of this seminar and the core of AI literacy
2. The **labour-versus-judgement line** and the **discovery-versus-verification distinction** are the two practical tools for applying that judgement in everyday research
3. **Sycophancy**, not hallucination, is the most dangerous failure mode in cognitive partnership; AI agreement is not evidence

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Questions and Discussion

Thank you!

Questions?

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